SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I **S.SHAMAH ALI FATHIMA(192573002)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**

- Discuss AI/ML cloud services, architecture, and industry applications.

8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
   - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
   - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **Server Virtualization and Its Role in Cloud Computing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. **Cloud Security: Threats and Protection Mechanisms**

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

    Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. **Software as a Service (SaaS): Architecture and Business Applications**

    Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title
- Aim and objective
- Core technical explanation
- Architecture or process diagram
- Methodology
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Topic : Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems

Introduction

- Internet of Things (IoT): network of connected sensors, devices, and actuators
- Cloud computing: on-demand storage, computing, and services over the internet
- IoT + cloud: collect, transmit, process, and visualize device data
- Enables remote monitoring, automation, and data-driven decisions

Architecture focus: device, edge/gateway, network, cloud platform, application

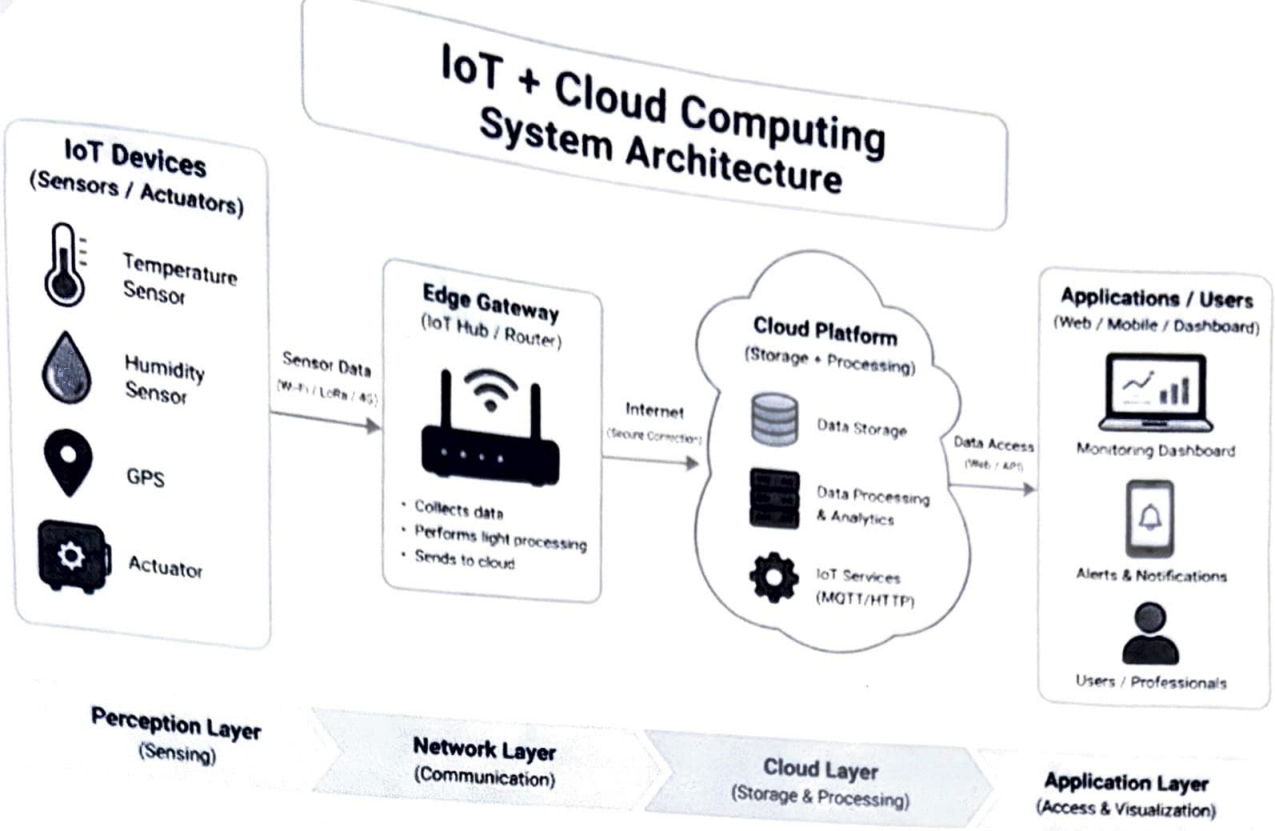
Objective

- Identify IoT and cloud architecture layers
- Explain device-to-cloud data flow
- Understand storage, processing, and analytics
- Highlight security, scalability, and real-world use

Core technical explanation:

- Large numbers of IoT devices generate continuous, heterogeneous data
- Device-only processing can be limited by compute, storage, and power
- Cloud integration introduces network dependence, latency, and security concerns
- Need: a layered architecture for reliable, scalable, secure device-to-cloud communication

Architecture or process diagram:



Real-world use case:

Smart agriculture: connected soil-moisture and weather sensors

Real-time entity: farm sensor network with cloud dashboard

- Sensors report soil moisture, temperature, and humidity
- Gateway forwards readings to a cloud IoT platform
- Cloud stores trends and compares readings with thresholds
- Dashboard supports remote irrigation monitoring
- Alerts can notify users when moisture falls below a set level

Illustrative use case; no specific deployment or measured results claimed

Conclusion and key takeaway:

IoT: Connects devices to the Internet.

Sensors: Collect data from devices.

Cloud: Stores and processes IoT data.

Internet: Transfers data between devices and cloud.

Benefits: Automation, real-time monitoring, scalability, and cost efficiency.

Applications: Healthcare, smart homes, industries, agriculture, and smart cities.

In short: IoT collects data → Cloud stores and processes it → Action is taken.

Photo of the presentation:



Conclusion:

IoT devices provide continuous sensing and actuation

- Cloud services add scalable storage, processing, and remote access
- Layered architecture clarifies data movement from devices to applications
- Edge processing can reduce unnecessary traffic and response delay
- Security, interoperability, and connectivity remain key design concerns
- Connected ecosystems support monitoring, automation, and informed decisions

: INTERNET OF THINGS [IOT] & CLOUD COMPUTING: BUILDING CONNECTED ECOSYSTEMS.

INTRODUCTION:

* Internet of Things (IOT) : network of connected sensors, devices & actuators.

* Cloud computing : on demand storage, computing & services over the internet.

* IOT + cloud : collect, transmit, process and visualize device data.

* Enables remote monitoring, automations & data-driven decisions.

OBJECTIVE :

* Identify IOT and cloud architecture layers.
* Explain device to cloud data flow
* Understand storage, processing and analytics.
* Highlight security, scalability & real world usage.

CORE - TECHNICAL EXPLANATION:

* Large numbers of IOT devices generate continuous, heterogeneous data.

* Device - only processing can be limited by compute, storage & power.

* Cloud integration introduces network dependance, latency & security concerns.

* Need: a layered architecture for reliable, scalable, secure device to cloud communication.